StudySapiens Solved Notes

Ch.3 Cells and Subcellular Organelles

Biology 11 • Punjab English Medium • PECTAA 2025–26 • Solved Exercise
Prompts are paraphrased and keyed to textbook question numbers. MCQ answer letters refer to the textbook options.

SECTION 1: MULTIPLE CHOICE QUESTIONS

Answer letters refer to the textbook options. Full textbook stems/options are not reproduced here.

1. Which eukaryotic cell structure does not contain DNA?

Answer: (c) Endoplasmic reticulum — Nucleus, mitochondria and chloroplasts contain DNA; the ER does not.

2. Which statement is not an accurate description of a chromosome?

Answer: (d) In eukaryotes, it is composed of many DNA molecules attached end to end — A eukaryotic chromosome normally contains one long linear DNA molecule associated with proteins, not many DNA molecules joined end to end.

3. What is a centriole?

Answer: (b) Composed of microtubules and important for organizing spindle fibres — Centrioles are cylindrical microtubule structures involved in organizing the spindle during cell division.

4. Which description fits rough endoplasmic reticulum?

Answer: (a) An intracellular membrane system to which ribosomes are attached — RER is a membrane network studded with ribosomes and is involved in protein synthesis/processing.

5. In the eukaryotic nucleus, genetic material complexed with protein is organized into what?

Answer: (c) Chromosomes — DNA plus histone and other proteins forms chromatin/chromosomes.

6. Which statement does not apply to the nuclear envelope?

Answer: (d) It has infoldings to form cristae — Cristae are infoldings of the inner mitochondrial membrane, not the nuclear envelope.

7. Lysosomes bud from which organelle?

Answer: (b) Golgi apparatus — Lysosomal enzymes are packaged by the Golgi apparatus into vesicles that form lysosomes.

studysapiens.com

8. Which function is carried out by peroxisomes?

Answer: (a) Break down fats and amino acids into smaller molecules usable for energy production — Peroxisomes carry out oxidative reactions, including fatty-acid breakdown, and produce hydrogen peroxide.

9. What is a likely effect of absent peroxisomes?

Answer: (b) Accumulation of hydrogen peroxide causing oxidative stress and cell damage — Peroxisomes contain catalase to break down H_2O_2; without them, toxic peroxide can accumulate.

10. Which statement does not apply to chloroplasts?

Answer: (d) They are bounded by two membranes, the inner folded into cristae — Chloroplasts have thylakoids/grana, not cristae; cristae belong to mitochondria.

11. Which sequence best represents the secretory protein pathway in the textbook?

Answer: (d) RER → SER → Golgi apparatus → cell membrane — The textbook pathway moves secretory proteins from RER through SER/Golgi processing to the cell membrane for release.

12. How does facilitated diffusion differ from active transport?

Answer: (b) Facilitated diffusion does not require energy; active transport does — Facilitated diffusion moves substances down a gradient via proteins; active transport uses energy to move against a gradient.

StudySapiens.com

SECTION 2: SHORT QUESTIONS

1. Compare the resolution and magnification of light and electron microscopes.

Light microscope	Electron microscope
Radiation: Visible light.	**Radiation:** Electron beam.
Useful magnification: Up to about ×1000–×1500.	**Useful magnification:** Up to about ×100,000–×500,000 or more.
Resolution: About 200 nm.	**Resolution:** About 0.2–2 nm (much higher).
Specimens: Living or stained cells/tissues.	**Specimens:** Usually dead, fixed, dehydrated specimens.

2. State the cell theory, how it can be validated, and its exceptions.

Cell theory states that all living organisms are made of cells, the cell is the basic unit of structure and function, and new cells arise from pre-existing cells. It is validated by microscopic observation of tissues, cell division and the continuity of cells in growth and reproduction.

Exceptions/limitations include viruses (acellular), striated muscle fibres and aseptate fungal hyphae (multinucleate), and mature mammalian red blood cells (no nucleus).

3. Complete the transport comparison table using YES/NO.

Simple diffusion	Facilitated diffusion	Active transport
Is ATP required?: NO	**Is ATP required?:** NO	**Is ATP required?:** YES
Are carrier proteins involved?: NO	**Are carrier proteins involved?:** YES	**Are carrier proteins involved?:** YES
Is direction always from higher to lower concentration?: YES	**Is direction always from higher to lower concentration?:** YES	**Is direction always from higher to lower concentration?:** NO

4. Categorize organelles as single-membrane, double-membrane or non-membranous.

Category	Organelles
Single membrane	ER, Golgi apparatus, lysosomes, peroxisomes, vacuoles, vesicles.
Double membrane	Nucleus, mitochondria, chloroplasts/plastids.
No membrane	Ribosomes, centrioles, cytoskeleton, nucleolus.

5. State two functions of plasma-membrane proteins.

Membrane proteins act as transport channels/carriers/pumps, and as receptors for signalling. They also function in cell recognition, adhesion and enzymatic reactions.

6. State two features mitochondria share with prokaryotes.

Mitochondria have their own circular DNA and 70S ribosomes. They also divide by a fission-like process and have double membranes, supporting their prokaryotic endosymbiotic origin.

7. List three differences between prokaryotic and eukaryotic cells.

Prokaryotic	Eukaryotic
Nucleus: No true nucleus; nucleoid.	**Nucleus:** True nucleus with nuclear envelope.
Organelles: No membrane-bound organelles.	**Organelles:** Membrane-bound organelles present.
DNA/ribosomes: Circular DNA; 70S ribosomes.	**DNA/ribosomes:** Linear chromosomes; 80S cytoplasmic ribosomes.

8. List structures/molecules that can cross the nuclear envelope.

Through nuclear pores: mRNA, tRNA and ribosomal subunits move out; proteins such as histones, polymerases and transcription factors move in. Small molecules and ions can also pass; the envelope regulates macromolecular traffic.

9. Distinguish the following pairs.

a. Exocytosis and endocytosis

Exocytosis	Endocytosis
Direction: Materials leave the cell.	**Direction:** Materials enter the cell.
Vesicle: Vesicle fuses with plasma membrane.	**Vesicle:** Plasma membrane invaginates to form vesicle.
Use: Secretion, waste removal, membrane addition.	**Use:** Uptake of nutrients, fluids, particles.

b. Phagocytosis and pinocytosis

Phagocytosis	Pinocytosis
Material: Large solid particles/cells.	**Material:** Extracellular fluid and dissolved solutes.
Vesicle: Large phagosome.	**Vesicle:** Small pinocytic vesicle.
Example: WBC engulfing bacteria.	**Example:** Non-specific fluid uptake by cells.

studysapiens.com

c. Peroxisome and glyoxysome

Peroxisome	Glyoxysome
Occurrence: Animals and plants.	**Occurrence:** Mainly germinating fat-rich seeds.
Main role: Oxidation of fatty acids; detoxification of H_2O_2.	**Main role:** Convert fats into carbohydrates via glyoxylate cycle.
Enzymes: Oxidases, catalase.	**Enzymes:** Glyoxylate-cycle enzymes plus peroxisomal enzymes.

10. What are the main functions of lysosomes?

Lysosomes digest macromolecules, worn-out organelles (autophagy) and engulfed particles (heterophagy). They recycle cellular components and can destroy the cell in autolysis.

11. Describe the role of the Golgi body in forming lysosomes.

Hydrolytic enzymes made in the RER are modified and packaged in the Golgi apparatus. The Golgi buds off vesicles containing these enzymes, which become primary lysosomes.

12. What are histones and where are they found?

Histones are basic proteins rich in lysine and arginine. They are found in the eukaryotic nucleus, where DNA wraps around them to form nucleosomes and chromatin.

13. What is a stem cell and what are its main uses?

Stem cells are unspecialized cells that can self-renew and differentiate into specialized cell types. Uses include regenerative medicine, treatment of damaged tissues, drug testing, disease modelling and personalized medicine.

14. Name structures A–G in the mitochondrion diagram.

Label	Structure
A	DNA (circular mitochondrial DNA)
B	Ribosomes (70S)
C	Matrix
D	Outer membrane
E	Inner membrane
F	F_1 particle (ATP synthase)
G	Cristae

studysapiens.com

The exercise asks A–G. The ribosome pointer is the second label (B); in the printed scan this letter is unclear and can look like H. The inter-membrane space is not separately lettered in this exercise diagram.

15. Label the cell electron micrograph and give evidence that it is metabolically active and secretory.

a. Main labelled parts

- Golgi apparatus; smooth ER; centriole; nucleus/nuclear envelope; nucleolus; lysosome/vesicle.
- Mitochondrion; rough ER; plasma membrane/cytoplasm.

b. Evidence of activity and secretion

The cell shows many mitochondria for ATP supply, extensive rough ER and ribosomes for protein synthesis, a prominent Golgi apparatus for processing/packaging, and secretory vesicles near the membrane. These features indicate active enzyme/protein secretion.

SECTION 3: LONG QUESTIONS

1. Describe the structure and chemical composition of eukaryotic and prokaryotic cell walls.

Prokaryotic cell wall

Bacterial walls are mainly peptidoglycan (murein): alternating N-acetylglucosamine (NAG) and N-acetylmuramic acid (NAM) chains cross-linked by short peptides. Gram-positive walls are thick and contain teichoic acids; Gram-negative walls are thin and covered by an outer membrane containing lipopolysaccharide.

Eukaryotic cell walls

- Plants: cellulose microfibrils embedded in a matrix of hemicellulose, pectin and sometimes lignin; primary wall is flexible, secondary wall may be thick and rigid.
- Fungi: chitin, a nitrogen-containing polysaccharide.
- Algae: cellulose, silica or other polysaccharides depending on the group.
- Animal cells: no cell wall; only a plasma membrane and extracellular matrix.

Thus wall chemistry is a key feature separating major groups of organisms.

2. Explain the chemical composition and functions of the plasma membrane.

The plasma membrane is a fluid mosaic of phospholipid bilayer with proteins, cholesterol and carbohydrates. Phospholipids have hydrophilic heads facing water and hydrophobic tails facing inward, forming a selective barrier.

- Proteins act as channels, carriers, pumps, receptors, enzymes and adhesion molecules.

studysapiens.com

- Cholesterol regulates fluidity and stability in animal cells.
- Carbohydrates attached to lipids/proteins form glycolipids/glycoproteins for recognition.

Functions

- Selective permeability and transport of ions/nutrients/wastes.
- Cell signalling through receptors and second messengers.
- Cell recognition, adhesion and tissue formation.
- Compartmentalization and maintenance of the internal environment.

3. Identify the role of glycolipids and glycoproteins as cell-surface markers.

Glycolipids and glycoproteins carry carbohydrate chains exposed on the outer surface of the plasma membrane. Together they form the glycocalyx, a sugar-rich coating that acts like an identification tag.

- They allow cells to recognize self from non-self, which is essential in immune responses.
- They determine blood-group antigens (e.g., ABO system).
- They help cells adhere to one another during tissue formation.
- They serve as receptors or co-receptors for signalling molecules and some pathogens.

Because the carbohydrate patterns differ between cell types, they provide molecular “name badges” for cell-cell recognition.

4. Explain the structure, chemical composition and function of ribosomes.

Ribosomes are non-membranous ribonucleoprotein particles made of ribosomal RNA (rRNA) and proteins. They consist of a large and a small subunit that join during protein synthesis.

Prokaryotic/organellar	Eukaryotic cytoplasmic
Sedimentation: 70S (50S + 30S).	**Sedimentation:** 80S (60S + 40S).
Location: Cytoplasm, mitochondria, chloroplasts.	**Location:** Cytoplasm and rough ER.

Function: ribosomes translate mRNA into polypeptides. The small subunit binds mRNA and tRNA anticodons; the large subunit catalyses peptide-bond formation. Free ribosomes make cytosolic proteins, while RER-bound ribosomes make secretory, membrane and lysosomal proteins.

studysapiens.com

5. Explain the structure and functions of the Golgi complex.

The Golgi complex consists of stacked, flattened membranous cisternae with a cis (forming) face near the ER and a trans (maturing) face near the plasma membrane, plus associated vesicles.

- Modifies proteins and lipids received from the ER, e.g., glycosylation and phosphorylation.
- Sorts and tags molecules for their correct destinations.
- Packages materials into secretory vesicles, lysosomes and transport vesicles.
- Synthesizes some polysaccharides and contributes to cell-wall material in plants.

It is especially prominent in secretory cells such as gland cells and plasma cells.

6. Describe the structure, chemical composition and function of chromosomes.

Eukaryotic chromosomes are linear DNA molecules complexed with histone and non-histone proteins. DNA wraps around histone octamers to form nucleosomes, which coil into chromatin fibres and condense into visible chromosomes during cell division.

- Chemical composition: DNA, histones (rich in lysine/arginine), non-histone scaffold proteins and small amounts of RNA.
- Each chromosome carries many genes arranged in a specific linear order.
- Centromeres attach spindle fibres; telomeres protect chromosome ends.

Functions: store genetic information, ensure accurate DNA replication and segregation, and regulate gene expression through chromatin structure.

studysapiens.com

7. Discuss the nuclear envelope and nuclear pore complex.

The nuclear envelope is a double membrane separating nucleoplasm from cytoplasm. The outer membrane is continuous with the rough ER and may bear ribosomes; the inner membrane is supported by the nuclear lamina. The perinuclear space lies between the two membranes.

Nuclear pore complexes perforate the envelope and regulate traffic: mRNA, tRNA and ribosomal subunits move out, while histones, polymerases, transcription factors and other nuclear proteins move in. Small molecules and ions can diffuse more freely.

Thus the envelope protects DNA and organizes chromatin, while pores control communication between nucleus and cytoplasm.

8. Explain how the Golgi apparatus is involved in making cell secretions.

> Protein synthesis on RER → Transport vesicle to cis-Golgi → Modification and sorting in Golgi cisternae → Packaging at trans-Golgi → Secretory vesicle → Exocytosis at plasma membrane

Proteins destined for secretion are synthesized on RER-bound ribosomes and enter the ER lumen. Transport vesicles carry them to the cis-Golgi. As they move through the cisternae, they are modified (e.g., glycosylation), concentrated and sorted. At the trans-Golgi network they are packaged into secretory vesicles, which fuse with the plasma membrane and release contents by exocytosis.

9. Describe the structure and functions of smooth and rough ER.

Rough ER	Smooth ER
Surface: Ribosomes attached; appears rough.	**Surface:** No ribosomes; appears smooth.
Structure: Flattened cisternae continuous with nuclear envelope.	**Structure:** Tubular network.
Main functions: Synthesis, folding and initial glycosylation of secretory/membrane proteins.	**Main functions:** Lipid/steroid synthesis, detoxification, carbohydrate metabolism, Ca^{2+} storage.
Abundance: Secretory cells such as pancreatic acinar cells.	**Abundance:** Liver, steroid-producing cells, muscle (sarcoplasmic reticulum).

Both are part of the endomembrane system and transport materials within the cell.

10. Explain the role of lysosomes and peroxisomes in regulating cellular contents.

Lysosomes contain acid hydrolases and digest macromolecules, worn-out organelles (autophagy) and engulfed particles (heterophagy). They recycle monomers back to the cytoplasm and can destroy the cell by autolysis when needed.

Peroxisomes contain oxidases that produce hydrogen peroxide and catalase that breaks H_2O_2 into water and oxygen. They oxidize fatty acids, detoxify alcohol and other harmful compounds, and participate in photorespiration in plants.

Together they control turnover of cellular components and prevent accumulation of toxic or unwanted materials.

11. Describe the three fibres of the cytoskeleton.

Fibre	Structure	Functions
Microfilaments	Actin polymers, about 7 nm diameter.	Cell shape, muscle contraction, cytoplasmic streaming, cytokinesis, cell movement.
Intermediate filaments	Rope-like proteins (e.g., keratins), about 10 nm.	Mechanical strength, anchoring organelles, maintaining cell shape.
Microtubules	Tubulin polymers, about 25 nm.	Tracks for organelle transport, spindle fibres, cilia/flagella, cell shape.

The cytoskeleton is dynamic: fibres assemble and disassemble to support movement, division and intracellular organization.

12. Describe the formation and functions of lysosomes.

Lysosomal enzymes are synthesized in the RER, transported to the Golgi apparatus, tagged (e.g., with mannose-6-phosphate) and packaged into vesicles that bud off as primary lysosomes.

Primary lysosomes fuse with endosomes, phagosomes or damaged organelles to form secondary lysosomes, where acid hydrolases digest the contents. Useful monomers are recycled; indigestible residue may be expelled.

- Intracellular digestion of macromolecules and particles.
- Autophagy: removal of worn-out organelles.
- Autolysis: programmed destruction of cells during development or injury.
- Defence: destruction of engulfed pathogens in phagocytes.

studysapiens.com

13. Compare mitochondria and chloroplasts as energy organelles.

Mitochondria	Chloroplasts
Energy conversion: Chemical energy of food → ATP (cellular respiration).	**Energy conversion:** Light energy → chemical energy of sugars (photosynthesis).
Internal membranes: Inner membrane folded into cristae.	**Internal membranes:** Thylakoids stacked into grana.
Fluid compartment: Matrix.	**Fluid compartment:** Stroma.
Pigments: No photosynthetic pigments.	**Pigments:** Chlorophyll and accessory pigments.
Own DNA/ribosomes: Circular DNA and 70S ribosomes.	**Own DNA/ribosomes:** Circular DNA and 70S ribosomes.
Gas exchange: Uses O_2, releases CO_2.	**Gas exchange:** Uses CO_2, releases O_2.

Both are double-membrane organelles with their own genetic machinery, supporting the endosymbiotic theory.

14. Describe the basic structure of a mitochondrion from outside inward.

- **Outer membrane:** smooth and permeable to small molecules due to porins.
- **Intermembrane space:** narrow compartment between outer and inner membranes; accumulates H^+ during respiration.
- **Inner membrane:** highly folded into cristae to increase surface area; contains electron transport chain and ATP synthase (F_1 particles).
- **Matrix:** gel-like interior containing enzymes for the Krebs cycle, mitochondrial DNA, RNA and 70S ribosomes.

The folding of the inner membrane maximizes ATP production, while the matrix houses the enzymes and genetic machinery needed for respiration.

15. Describe the pathway of protein and steroid signals from outside to inside a cell.

Protein/peptide signal

> Signal binds membrane receptor → Second messenger cascade → Protein kinases activated → Cellular response / gene expression

Steroid signal

> Lipid-soluble hormone crosses membrane → Binds intracellular receptor → Hormone-receptor complex enters nucleus → Binds DNA and regulates transcription

Protein signals are water-soluble and cannot cross the lipid bilayer, so they act through surface receptors and second messengers. Steroid signals are lipid-soluble, cross the membrane and act through internal receptors that directly regulate gene expression.

16. Categorize and explain different types of stem cells.

Type	Source/potency	Notes
Embryonic stem cells	Inner cell mass of blastocyst; pluripotent.	Can form almost all body cell types; ethical issues.
Adult stem cells	Bone marrow, fat, blood, tissues; multipotent.	Limited differentiation; lower rejection risk.
Induced pluripotent stem cells	Reprogrammed adult somatic cells; pluripotent.	Avoid embryo use but may cause genetic changes/tumours.

By potency: totipotent cells can form all cell types including extra-embryonic tissues; pluripotent cells form almost all body cells; multipotent cells form related cell families; unipotent cells produce one cell type.

17. What are the advantages and disadvantages of induced pluripotent stem cells?

Advantages	Disadvantages
No destruction of embryos.	Reprogramming may introduce genetic changes.
Can be patient-specific, reducing immune rejection.	Risk of tumour/teratoma formation.
Pluripotent for regenerative medicine and disease modelling.	Difficult to direct differentiation into fully functional cells.
Useful for drug testing and personalized medicine.	Ethical/safety concerns and high cost.

iPSCs are produced by introducing reprogramming factors into adult somatic cells, returning them to an embryonic-like pluripotent state.

INQUISITIVE QUESTIONS

1. What can be inferred if a cell has an exceptionally large Golgi apparatus?

The cell is likely highly active in secretion, processing and packaging proteins or lipids. Examples include gland cells, plasma cells and other secretory cells.

2. How does a lipid-soluble steroid hormone most likely act inside a target cell?

It diffuses through the plasma membrane and binds an intracellular receptor. The hormone-receptor complex acts as a transcription regulator in the nucleus, changing gene expression.

3. Why are endocytosis and exocytosis categorized as active transport?

They require metabolic energy (ATP) for membrane deformation, vesicle formation, movement and fusion. They move large particles or bulk material, not simple diffusion through the bilayer.

4. Justify why the membrane is described as fluid.

Phospholipids and many proteins can move laterally within the bilayer. The membrane is flexible, self-sealing and dynamic; fluidity is influenced by temperature, fatty-acid saturation and cholesterol.